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 Studierichting: Informatica
 Oefeningenreeks 6E nr. 1.5

$$T_7(f, 2)(x) \text{ voor } f(x) = \frac{1}{x-1}$$

$$f(x) = (x-1)^{-1}$$

$$f'(x) = -1(x-1)^{-2}$$

$$f''(x) = 2(x-1)^{-3}$$

$$f'''(x) = -6(x-1)^{-4}$$

$$f^{iv}(x) = 24(x-1)^{-5}$$

$$f^v(x) = -120(x-1)^{-6}$$

$$f^{vi}(x) = 720(x-1)^{-7}$$

$$f^{vii}(x) = -5040(x-1)^{-8}$$

$$f(2) = 1$$

$$f'(2) = -1$$

$$f''(2) = 2$$

$$f'''(2) = -6$$

$$f^{iv}(2) = 24$$

$$f^v(2) = -120$$

$$f^{vi}(2) = 720$$

$$f^{vii}(2) = -5040$$

$$\begin{aligned} \Rightarrow T_7(f, 2)(x) &= \sum_{k=0}^7 \frac{f^{(k)}(2)}{k!} (x-2)^k \\ &= \frac{1}{0!} + \frac{-1}{1!}(x-2) + \frac{2}{2!}(x-2)^2 + \frac{-6}{3!}(x-2)^3 + \frac{24}{4!}(x-2)^4 \\ &\quad + \frac{-120}{5!}(x-2)^5 + \frac{720}{6!}(x-2)^6 + \frac{-5040}{7!}(x-2)^7 \\ &= 1 - (x-2) + (x-2)^2 - (x-2)^3 + (x-2)^4 - (x-2)^5 + (x-2)^6 - (x-2)^7 \end{aligned}$$

References